

- 6 Fig. 6.1 shows a long solenoid which has a small copper disc mounted at its centre.

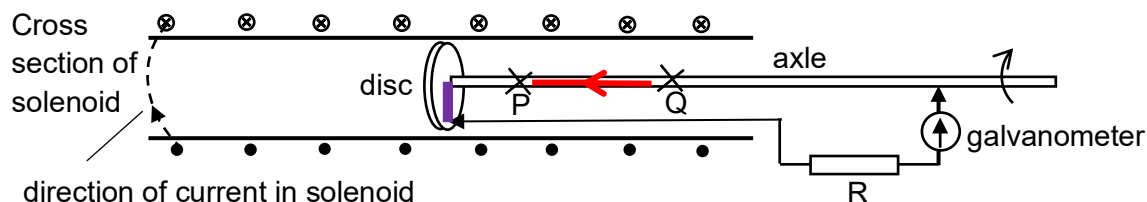


Fig. 6.1

The disc spins on a conducting axle which lies along the axis of the solenoid. The solenoid is connected in series with a d.c. supply and the direction of the current in the solenoid is as shown in Fig. 6.1. One end of the resistor is connected to the rim of the disc through brushes. The other end is connected to a galvanometer.

- (a) (i) Indicate the direction of the induced current between point P and Q with an arrow on Fig. 6.1. [1]
- (ii) Explain why an e.m.f. is generated between the axle and the rim of the disc when the disc rotates.

The solenoid sets up a uniform magnetic field perpendicular to the plane of the disc. When the disc turns, there is a continuous flux cutting across the metal strip from centre of the disc to rim. By Faraday's law, this results in induced e.m.f. across this strip from centre of the disc to rim as drawn the diagram.

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.....[3]

- (b) State the effect on the galvanometer, if any, when

- (i) the speed of the rotation of the disc is increased,

Larger angle of deflection.

.....[1]

- (ii) the direction of the rotation of the disc is changed.

Same angle of deflection in the other direction.

.....[1]

- (c) A voltmeter is now connected across the rim of the disc. Explain why there is no reading.

The rate of flux cutting is the same for any strip of conductor from the centre to the rim, thus, the rim will be at the same potential, the potential difference across 2 points of the rim will be zero and hence no reading.

.....[2]

[Total: 8]